Edward Hirst

0421 969 501 | Melbourne, VIC | edthirst@gmail.com | github.com/tikitaik | linkedin.com/in/ed-hirst

Education

The University of Melbourne **March 2025 – November 2027 (Expected)**
Bachelor of Science in Computing and Software Systems Parkville, VIC

- H1 WAM (82.27)
- Relevant coursework: Algorithms & Data Structures (COMP20007), Object Oriented Software Development (SWEN20003), Database Systems (INFO20003)

Northcote High School **January 2019 – December 2024**
Victorian Certificate of Education Northcote, VIC

- 98.90 ATAR, Algorithmics (Higher Education Study Subject) & Chemistry Dux
- Sir Gustav Nossal Award (2024): General science achievement award

Projects

Self-Hosted Server | GitHub | Arch Linux, Bash, Cloudflare, Apache, Java March 2026 – Present

- Self-hosted headless Arch Linux installation accessed through SSH
- Utilises Cloudflare Tunnels to expose website to public traffic without opening inbound ports
- Hosts a Minecraft server with Spigot plugins, allowing concurrent Java and Bedrock edition client access
- Integrated Apache to serve a static HTML page with URL rewrites implemented
- Updates Cloudflare's DNS records automatically leveraging Cloudflare's API and cron to enable dynamic IP

3D Physically-Based Renderer | GitHub | C++, OpenGL, CMake April 2025 – February 2026

- Created a physically-based 3D rendering engine with OpenGL, C++, and CMake
- Implemented graphics algorithms including PBR and Blinn-Phong lighting, screen-space ambient occlusion, normal/parallax mapping, instancing, and model loading
- Employed CMake to enable streamlined program building across Windows, Mac, and Linux

Experience

Team Member & Community Donations Centre Ambassador **October 2024 – Present**
Savers Brunswick, VIC

- Operated community donation centre with and without a partner on weekends, requiring accepting, stacking, and pricing donations, processing up to 4 tonnes of product per day
- Trained 5 new employees in retail/community donation centre skills
- Learned new POS system from the handbook and helped supervisors and managers use it
- Collaborated in a team of up to 15 to operate checkouts, maintain store cleanliness, and sort clothing
- Communicated clearly and politely with people from diverse backgrounds to ensure customer satisfaction

Volunteer Assistant Coach **May 2021 – November 2022**
Brunswick Zebras F.C. Brunswick, VIC

- Assisted coaches in running 1.5 hour Saturday morning soccer training sessions for up to 100 children aged 4–8
- Led individual groups of 20 in warmups, drills, and matches, mitigating disputes and conflicts between children
- Collaborated with 3 assistant coaches to set up and pack down equipment efficiently and neatly

Web Development Intern (Work Experience) **May 2022**
Medtasker Melbourne, VIC

- Developed an HTML/CSS webpage to serve as Medtasker's support interface for the Tasmanian Health Service
- Assisted live deployment to the Tasmanian Health Service's systems
- Renewed Medtasker's website's SSL certificate and managed company contacts using HubSpot

Technical Capabilities

- Languages: C/C++, Bash, Java, Python, HTML/CSS
- Tools: Linux/UNIX CLI, Git, SSH, Apache, Cloudflare, CMake, OpenGL, Visual Studio
- Skills: Linux system administration, graphics programming/pipeline, networking, build systems

Interests

- Playing music
- Technology
- Reading